INSTALLATION AND ACCEPTANCE TESTING OF NETWORK, ANTENNA COMPENSATION, CN-1404/ARN-89, IN ARMY AIRCRAFT

This specification is approved for use by Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification provides installation, equipment performance, engineering and production approval instructions and requirements for installation and testing of Network, Antenna Compensation, CN-1404/ARN-89, and associated equipment in prototype and production model aircraft.

2. APPLICABLE DOCUMENTS

2.1 The following Government documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-B-5087	Bonding, Electrical, and Lightning Protection, for Aerospace Systems.
MIL-W-5088	Wiring, Aircraft, Selection and Installation of.
MIL-E-5400	Electronic Equipment, Airborne, General Specification for.
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment,
MIL-E-6051	Electromagnetic Compatibility Requirements, Systems.
MIL-I-8700	Installation and Test of Electronic Equipment in Aircraft, General Specification for.
MIL-C-45662	Calibration System Requirements.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Electronics Command, ATTN: DRSEL-RD-TS-S, Fort Monmouth, New Jersey 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-463	Definitions and Systems of Units, Electromagnetic Interference Technology.
MIL-STD-704	Aircraft Electric Power Characteristics.

PUBLICATIONS

TECHNICAL INSTRUCTIONS

TECHNICAL INSTRUCTIONS

SCL-T-0049	Avionics Bench, Preflight and Flight Test Instructions for Direction Finder Set AN/ARN-89( ).
SCL-I-0049	Avionics Installation Instructions for Direction Finder Set AN/ARN-89( ).

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 General

3.1.1 Acceptance. - Any installation of the CN-1404/ARN-89 which is in accordance with MIL-I-8700, Technical Instruction SCL-I-0049 (for the AN/ARN-89( )), and the requirements of this specification shall be considered acceptable. Such acceptance, however, does not constitute safety of flight approval.

3.1.2 Responsibility for Miscellaneous Components. - The contractor shall furnish and install all required plugs, cables, brackets, connectors, and miscellaneous items to complete satisfactory installation and operation of the CN-1404/ARN-89. All contractor furnished components shall be in accordance with the applicable electromagnetic interference and compatibility requirements in MIL-STD-461 and the requirements in MIL-E-5400 for Class I-A equipment designed for operation in both aircraft and helicopters.

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3.1.3 Quality Assurance. - The test and inspection of the CN-1404/ARN-89 installation shall be in accordance with the procedures set forth in MIL-I-8700 and Technical Instruction SCL-T-0049 and Section 4 of this specification.

3.1.4 Check-off List. - The contractor shall record the results of all tests conducted on each equipment on a check-off list. The contractor shall prepare a check-off list for each CN-1404/ARN-89 installed in an aircraft and maintain a file of these records for reference, as necessary, by the procuring activity.

3.1.5 Equipment Failure. - In the case of equipment failure the requirements and procedures specified in MIL-I-8700 shall govern with the exception that replacement of component parts shall be made only upon specific written approval by the procuring activity. The applicable "Unserviceable Materiel Report" shall be utilized when reporting equipment failure. Shipping instructions for the faulty equipment and distribution requirements for the equipment failure report shall be specified by the contracting officer.

3.1.6 Procuring Activity Approval. - In all cases where approval by the procuring activity is specified, the contractor shall be in receipt of said approval prior to conducting any tests, inspections, or otherwise utilizing the proposed equipment or procedures.

3.1.7 Test Equipment Calibration. - All test equipment calibration shall conform to MIL-C-45662, Calibration System Requirements. The contractor shall prepare a statement which identifies all test equipment to be utilized and certifies the calibration of this test equipment in accordance with MIL-C-45662. This certification shall be forwarded to the procuring activity as a supplement to the test report.

3.1.8 Electromagnetic Compatibility. - Interference control and test plans shall be prepared by the contractor in accordance with MIL-E-6051 and MIL-STD-463 and submitted to the procuring activity for approval. After receipts of approval, the contractor shall conduct the tests specified in 4.3 to demonstrate compliance with MIL-E-6051 and to insure interference free operation of the CN-1404/ARN-89 after installation in an aircraft. A contractor prepared electromagnetic compatibility test report shall be forwarded to the procuring activity as directed by the contracting officer. EMI/EMC characteristics of the CN-1404/ARN-89 which verify compliance with the applicable requirements of MIL-STD-461 will be provided by the procuring activity as required.

3.1.9 Equipment To Be Tested. - The equipment listed in Table I shall meet the requirements specified herein. Whenever the model or detail specification for a particular aircraft provides a list not in agreement with Table I, the list in the aircraft specification shall govern.

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TABLE I

ELECTRONIC EQUIPMENT TO BE TESTED FOR
NETWORK, ANTENNA COMPENSATION, CN-1404/ARN-89

Equipment Designation	Part No. or Nomenclature	No. Of Units Required	Wt. Per Unit (Typical)	Unit Heat Dissipation (Max.)
Network, Antenna Compensation	CN-1404/ARN-89	1	0.5 pounds	1.2W

3.1.10 Additional Equipment Required. - In addition to the equipment listed in Table I, the following equipment listed in Table II, is required for complete installation of the Network, Antenna Compensation, CN-1404/ARN-89.

TABLE II

ELECTRONIC EQUIPMENT ASSEMBLY TABLE
FOR NETWORK, ANTENNA COMPENSATION, CN-1404/ARN-89

Equipment Designation	Part No. or Nomenclature	No. of Units Required	Remarks
Receiver	R1496/ARN-89	1	*
Control	C7392/ARN-89	1	*
Sense Pre-Amp	AM-4895( )/ARN-89	1	*
Loop Antenna	AS-2108/ARN-89	1	*
Sense Antenna	-	1	**

*Part of Direction Finder Set, AN/ARN-89( )

**Compatible antenna design will be specified in the appropriate aircraft detail specification.

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3.1.11 Test Equipment Required. - The following test equipment or equipment of equal or superior characteristics, shall be used in the performance of the tests specified herein. Items which are considered unique to the equipment under test and which may be loaned to the aircraft contractor by the Electronics Command are marked with an asterisk.

TABLE III

TEST EQUIPMENT AND ACCESSORIES

NAME	PURPOSE	PREFERRED TYPE	
		Type Designation	Characteristics
Test Kit	Interconnections & Terminations	MK-994/AR*	-
Signal Generator	Test Signal	AN/URM-25D	Freq. 1-50MHz Atten Rg 0-132db
Voltmeter	AC RMS Measurements	ME-30A/U	0.5-300 VAC
Multimeter	Check Voltage Current Levels	TS-352B/U	min: 0 - 30 VAC min: 0 - 30 VDC
Frequency Counter	Monitor Signal	AN/USM-307	0-10 MHz
Headset	Provides/Mounters Audio	H-101A/U	-
Stop Watch	Time Bearings	-	0.1 sec accuracy

3.2 Installation. - The installation of the CN-1404/ARN-89 (See Figure 1) shall conform with the general requirements of Specification MIL-I-8700 and Figure 2, CN-1404/ARN-89 outline dimensions.

3.2.1 Precautions. - The following precautions shall be observed when installing the CN-1404/ARN-89.

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a. Avoid sharp bends in all cables wherever possible.

b. Avoid running antenna interconnecting cables in cable ducts carrying aircraft or rf power cables.

c. The CN-1404/ARN-89 shall be bonded to ground in such a manner as to provide the shortest possible dc path to ground. Bonding shall be in accordance with MIL-B-5087.

d. Do not connect, disconnect, remove or replace the CN-1404/ARN-89 while power is applied to the ADF system.

e. Continuity, short, and grounding checks on the loop antenna and CN-1404/ARN-89 interconnecting cables shall be made prior to equipment connections.

f. Caution shall be exercised to assure that the antenna interconnection cables and the receiver interconnection cables are connected to the input and output of the CN-1404/ARN-89 respectively.

g. Check polarity of dc supply prior to connection to any unit.

h. Care shall be taken to prevent entrance of foreign matter into either the equipment, or equipment compartments.

CAUTION: Loop antenna cables and connectors have +15V dc on the center conductor of the coaxial cable. DO NOT short center conductor to ground or serious damage to the equipment can result.

3.2.2 Mounting Instructions. - (See Figure 3)

a. Position CN-1404/ARN-89 in accordance with instructions for the aircraft in which it is to be installed, and the general guidelines of this specification, so that the compensation adjustment is accessible.

b. Spot and drill fixing holes in accordance with the dimensional data shown in Figure 2.

c. Attach the unit to its mounting surface.

d. Disconnect cable plugs 4P1 and 4P2 from 4J1 and 4J2 on the loop antenna AS-2108( )/ARN-89.

e. Connect cable plugs 5P4 and 5P3 to 4J1 and 4J2 on the loop antenna AS-2108( )/ARN-89.

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f. Connect cable plugs 5P1 and 5P2 to 5J1 and 5J2 on the CN-1404/ARN-89.

NOTE: Steps e and f will install the contractor-furnished cable assemblies, if applicable.

g. Connect cable plugs 4P1 and 4P2 to 5J4 and 5J3 on the CN-1404/ARN-89. This will reconnect the cable plugs of the existing cable assemblies, signal, Y-loop and signal, and X-loop, completing the installation.

h. Verify that the goniometer is adjusted to zero compensation.

3.2.3 Cabling and Connectors. - In addition to the cables necessary for operation of the AN/ARN-89( ), two additional cables are required for the CN-1404/ARN-89. Typical cable assemblies are shown in figures 4 and 5. Note: Cable between the receiver and the CN-1404/ARN-89 are not interchangeable with those between the CN-1404/ARN-89 and the loop antenna because of different connectors.

3.3 Input Requirements.

3.3.1 Electrical Power. - The CN-1404/ARN-89 has been designed for reliable performance in operation when electrical input power, having characteristics as specified, and as defined in Standard MIL-STD-704, is applied to the AN/ARN-89( ).

3.3.2 DC Power. - The power requirements of the CN-1404/ARN-89 are supplied by and are included in the DC power envelope of the AN/ARN-89( ).

3.3.3 Environmental Requirements. The CN-1404/ARN-89 complies with the environmental requirements for Class I equipment designed for operation in both helicopters and aircraft as specified in MIL-E-5400 where indicated by a "C" (compliance) in Tables IV-A (operating) and IV-B (non-operating). The environmental limits marked "NC"(non-compliance) are specified in 3.3.7.The CN-1404/ARN-89 shall be installed within the limits specified.

3.3.4 Equipment Total Weight. - The CN-1404/ARN-89, excluding interconnecting cables, shall not weigh more than 0.5 pounds maximum.

3.3.5 Equipment Heat Dissipation. - Total heat dissipation is as follows:

Condition	Watts
Normal Operation	1.2 maximum

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3.3.6 Location Considerations. - Every consideration shall be given in the location of the CN-1404/ARN-89 and in installation requirements to promote ease of adjustment and maintenance. The considerations shall not be compromised to facilitate aircraft production techniques. The location details for the CN-1404/ARN-89 are general in nature only and are intended to provide the aircraft manufacturer (or the organization designing the installation of equipment) with pertinent points to be considered in developing the final installation of the CN-1404/ARN-89 in the aircraft. The preferred mounting location of the CN-1404/ARN-89 is within two (2.0) feet of the loop antenna. If, however, this is not feasible, the CN-1404/ARN-89 may be mounted in any location such that the total receiver to loop antenna cabling is not increased by more than (2.0) feet. In this event alternate cables will have to be fabricated.

3.3.7 Detailed Requirements. - The CN-1404/ARN-89 meets the environmental requirements for Class I equipment designed for operation in helicopters and aircraft as specified in MIL-E-5400. The following limits apply:

Temperature:	(degrees °F)	
Operating	-25°F to +160°F	
	-65°F to -25°F	Measurements may decrease by factor of 2
Non-Operating	-65°F to +160°F	
Altitude:	50,000 feet	
Humidity:	95%	
Shock:		
Operational	15g's	
Non-Operational	30g's	

Vibration: (Normally Mounted): Curve III Figure 2 of MIL-T-5422.

Frequency Hz	
5-20	0.1 in. double amplitude
20-33	2g's peak
33-55	0.036 in. double amplitude
55-550	5g's peak

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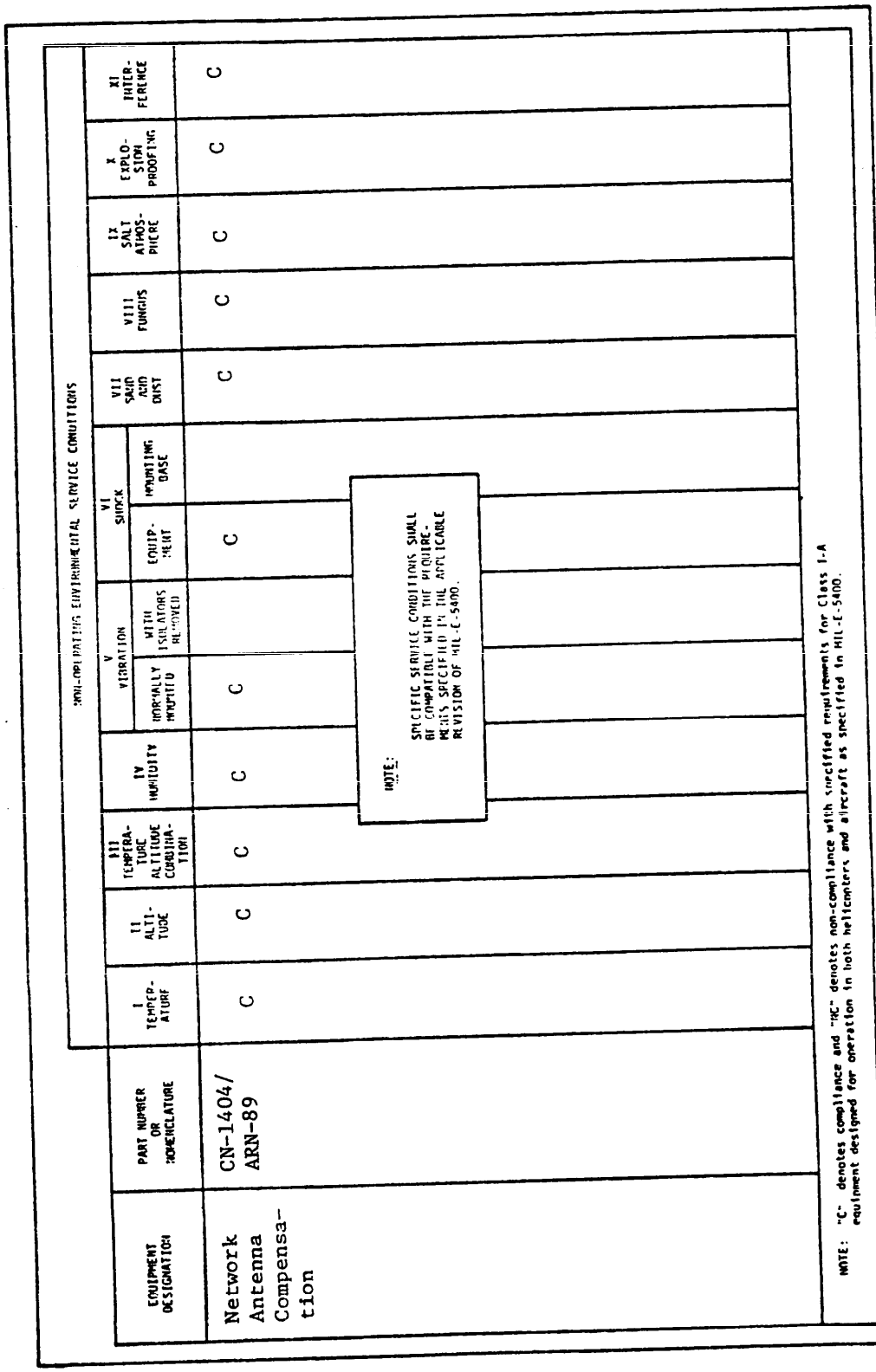
EQUIPMENT DESIGNATION	PART NUMBER OR NOMENCLATURE	OPERATING ENVIRONMENTAL SERVICE CONDITIONS												
		I TEMPERATURE	II ALTITUDE	III TEMPERATURE ALTITUDE COMBINATION	IV HUMIDITY	V VIBRATION		VI SHOCK		VII SAND AND DUST	VIII FUNGUS	IX SALT ATMOSPHERE	X EXPLOSION PROOFING	XI INTERFERENCE
						ASSEMBLY MOUNTED	WITH ISOLATORS REMOVED	EQUIPMENT	MOUNTING BASE					
Network, Antenna Compensation	CN-1404/ ARN-89	C	C	C	C	C		C		C	C	C	C	C

NOTE: SPECIFIC SERVICE CONDITIONS SHALL BE COMPATIBLE WITH THE REQUIREMENTS SPECIFIED IN THE APPLICABLE REVISION OF MIL-E-5400.

NOTE: "C" denotes compliance and "NC" denotes non-compliance with specified requirements for Class 1-A equipment designed for operation in both helicopters and aircraft as specified in MIL-E-5400.

TABLE IV - A

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EQUIPMENT DESIGNATION	PART NUMBER OR NOMENCLATURE	NON-OPERATING ENVIRONMENTAL SERVICE CONDITIONS												
		I TEMPERATURE	II ALTITUDE	III TEMPERATURE ALTITUDE COMBINATION	IV HUMIDITY	V VIBRATION		VI SHOCK		VII SAND AND DUST	VIII FUNGUS	IX SALT ATMOSPHERE	X EXPLOSION PROOFING	XI INTERFERENCE
						NORMALLY MOUNTED	WITH ISOLATORS REMOVED	EQUIPMENT	MOUNTING BASE					
Network Antenna Compensation	CN-1404/ ARN-89	C	C	C	C	C		C		C	C	C	C	C

NOTE: SPECIFIC SERVICE CONDITIONS SHALL BE COMPATIBLE WITH THE REQUIREMENTS SPECIFIED IN THIL APPLICABLE REVISION OF MIL-E-5400.

NOTE: "C" denotes compliance and "NC" denotes non-compliance with specified requirements for Class 1-A equipment designed for operation in both helicopters and aircraft as specified in MIL-E-5400.

TABLE IV - B

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3.4 Bench Tests. - Bench tests specified in paragraph 4.4 are required to determine that the equipment operates satisfactorily.

3.5 Prototype Flight Tests. - These tests shall be conducted at the point of initial installation on the first completed aircraft of a production contract unless otherwise directed by the procuring agency. Any time that there is a major change in aircraft configuration or equipment installation design, retesting is required on the first so modified aircraft. The tests shall be conducted to insure a satisfactory installation design and shall consist of preflight tests. These tests are designed to demonstrate that Direction Finder Set AN/ARN-89( ) with the CN-1404/ARN-89 installed, will perform in a manner comensurate with its design capabilities and will be compatible with the aircraft and the other items of the aircraft's electrical and electronic system when operated in flight.

3.5.1 Preflight Tests. - Preflight tests as specified in Paragraphs 4.3.1.1 to 4.3.1.6 of SCL-T-0049 and Paragraph 4.5 of this specification are to be performed prior to the flight tests of Paragraph 4.3.2 of SCL-T-0049 and after the bench tests of paragraph 4.4 of this specification to insure that the equipment has been installed properly and that the equipment as installed will meet the applicable requirements of this specification and referenced specifications. EMC tests shall be performed in accordance with MIL-E-6051 and a Government approved test plan.

3.6 Production Flight Tests. - Production flight tests are to be performed on every aircraft submitted for acceptance to determine that the CN-1404/ARN-89, installed in an AN/ARN-89( ) System, is operating properly. The production flight tests shall be the same as the prototype flight tests of Paragraphs 4.6 through 4.6.10.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of Examinations and Tests. - The examinations and tests required to assure conformance with the requirements of Section 3 are classified as follows:

a. Electromagnetic compatibility tests.

b. Installation.

c. Bench tests.

d. Prototype approval tests.

e. Production approval tests.

4.3 Electromagnetic Compatibility Tests. - Under the test conditions and procedures specified in the USAECOM approved, contractor prepared EMC control and EMC test plans, demonstrate compliance with 3.1.8.

4.4 Bench Tests

4.4.1 General. - A test bench is required, cabled to accommodate the units to be tested and provided with the various units of test equipment needed for testing (see Paragraph 3.1.11). The test facilities kit MK-99-/AR will be used to connect the units under test. The cabling between units shall be in accordance with Specification MIL-W-5088 and a typical aircraft wiring diagram and shall be so arranged that the component parts are accessible. The power source should be connected into the test bench wiring and should be protected by suitable fuses or circuit breakers. Allow required warmup time for each piece of test equipment.
CAUTION: Loop and sense antenna cables and connectors have +15 vdc on center conductor of coaxial cable. DO NOT short center conductor to ground or serious damage to the equipment can result.

4.4.2 Standard Conditions. - Standard conditions as hereafter referred to, will be as follows:

a. Input voltage: 27.5 $\pm$ 0.5 volts dc

b. Temperature: Room ambient (25°C $\pm$ 5°C)

c. Altitude: Normal Ground

d. Humidity: Room ambient up to 90% relative humidity

e. Vibration: None

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4.4.3 External Visual Inspection. - Inspect each CN-1404/ARN-89 externally for evidence of shipping and handling damage. Check all controls and mechnical assemblies for loose or missing screws, bolts and nuts. Inspect all connectors for looseness, damage, or missing parts. Check that the switch rotates freely without binding or excessive looseness.

4.4.4 Test Procedure. - Using MK-994/AR, connect an CN-1404/ARN-89 and an AN/ARN-89( ) to the test equipment as shown in Figure 6.

4.4.4.1 Preliminary ADF System Tests

4.4.4.1.1 Compensation Network Adjustment. - The CN-1404/ARN-89 adjustment switch must be set to postion "0" for the following tests.

4.4.4.1.2 ADF Tests. - Perform the tests of paragraphs 4.2.4 through 4.2.4.1.8 of SCL-T-0049, with special attention given to paragraph 4.2.4.1.4g. of SCL-T-0049.

4.4.4.1.3 Test Facilities Kit Control Positions. -

Control	Position
ADF Functional Selector Swtich	COMP
ADF Test Selector Switch	5
COMM CONT No. 1 Vol Control	Comfortable Listening Level
COMM CONT No. 1 NAV Toggle Switch	ON
DC POWER Circuit Breaker	ON

4.4.4.1.4 Compensation Adjustment Test. -

a. On the radio set control, place COMP-ANT-LOOP selector switch to COMP position.

b. Set the KILOHERTZ dial to the 0120 KHz setting.

c. Adjust the AN/URM-25D to give 50 uV output with no modulation. At 120 KHz, fine tune for maximum indication of the TUNE meter.

d. Observe HOMING/ADF indicator which should be resting at 45° $\pm$ 5° from top of indicator.

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e. Adjust the CN-1404/ARN-89 compensation adjustment (slotted screwdriver adjustment) to the max. clockwise position. The HOMING/ADF indicator should now be resting at 65° ± 5° from top of the indicator.

f. Adjust the CN-1404/ARN-89 compensation adjustment back toward zero compensation. Observe that each switch position reduces the HOMING/ADF indicator needle indication by approximately 2°.

NOTE: The AN/ARN-89( ) goniometer shall be set for zero compensation and the CN-1404/ARN-89 compensation selector shall be set for the appropriate aircraft setting, during preflight tests.

4.5 Preflight Test.

4.5.1 Test Equipment and Conditions. - The primary radio range stations used for navigation are in the 200-400 KHz frequency range; therefore, all adjustments shall be made using stations in this frequency range to assure optimum performance for navigation. Do not collect data during the period starting two hours before sunset and ending two hours after sunrise.

For preflight tests, select two or three radio range stations in the 200-400 KHz range which have different bearings from the aircraft and are at different distances between 10 miles and 100 miles.

To allow accurate assessment of bearing indications, the aircraft must be located not less than 200 feet from electrically conducting objects. This includes metallic frame buildings, hills, power lines and railroads.

NOTE: Performance of the ground test procedures may prove extremely difficult, as finding a location free of large structures, structural steel buildings, or steel imbedded in concrete and free of radio interference is remote. All of the mentioned items will contribute to system error which can only be properly corrected by a flight test.

4.5.2 Compensation Adjustment Ground Check. -

a. Swing aircraft heading to point exactly toward a selected radio range station.

NOTE: Accurate heading is necessary and accomplished by laying out a line on the ground using surveying instruments or by using a calibrated aircraft gyro compass.

b. Place COMP-ANT-LOOP selector switch in the COMP mode.

c. The bearing indicator will point at the radio station. Indicated bearing must be zero degrees ±3°.

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NOTE: Accuracy of this indication is dependent upon exactness obtained in step a. If goniometer compensator has not been adjusted, the error may exceed ±3°.

d. Swing aircraft heading 15° to the right of the original heading. Bearing indicator must remain aligned with selected station heading and should incidate a bearing of 345 ± 3°. Incorrect bearing indication shows that goniometer compensator requires adjustment.

NOTE: If bearing indicator pointer swings right, check for improper connection of bearing indicator.

e. Swing aircraft heading 30° to the right of the previous heading. Repeat until a full 360° swing is accomplished in 30° steps. After each 30° swing, note the bearing indicator readings. If the error exceeds ±3° refer to para 4.6.7 for CN-1404/ARN-89 adjustment.

4.6 Prototype Approval Tests. - Prototype approval tests are required for final acceptance of the equipment installation and shall be conducted on a prototype aircraft installation or on the first aircraft of a given contract. These tests are also required on the first aircraft following a major change, either in the aircraft or in the equipment installation design. These tests shall consist of both preflight tests of Paragraph 4.5 of this specification and the flight tests of para 4.3.2 of SCL-T-0049.

4.6.1 Electromagnetic Compatibility Tests (EMC). - EMC tests must be run in accordance with the Government approved test plan as required in Paragraph 3.3 of MIL-E-6051.

4.6.2 Test Equipment and Conditions. - For all flight tests, fly at 1000 - 1200 ft. altitude in level flight and select two or three radio range stations in the 200 - 400 KHz range which have different bearings from the aircraft and at different distances between 10 miles and 100 miles. Do not conduct testing during the period, starting 2 hours before sunset, and ending 2 hours after sunrise. Accurately determine the magnetic bearings, at the range radio station, to the flight reference point.

4.6.3 Loop Mode. -

a. Place COMP-ANT-LOOP selector switch in the LOOP position.

b. While flying a steady course, adjust LOOP L-R switch to obtain a null on the selected station. Note bearing indicator indication.

c. Adjust LOOP L-R switch to rotate bearing indicator and obtain opposite null. Note bearing indicator indication.

NOTE: Indicated bearing should be approximately 180° displaced from the original bearing obtained in Step b.

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d. Adjust LOOP L-R switch to rotate bearing indicator until signal is maximum on TUNE meter (approximately 90° from bearing obtained in either Step b or Step c). Be certain to remain below the agc threshold. Adjust the AUDIO control until TUNE meter indicates to the third scale division.

e. Place CW-VOICE-TEST toggle switch in the CW position.

f. Check for a detectable bfo tone in the audio system.

4.6.4 Sense Antenna Setup Check. -

a. Establish loop mode of operation using procedure of Paragraph 4.6.3, Steps a through d.

b. Place COMP-ANT-LOOP selector switch in the ANT position.

c. Note that TUNE meter indication is between 2.5 and 3.5 scale divisions. If not, note difference and adjust Impedance Matching Amplifier AM-4859( )/ARN-89 to compensate for the difference. Recheck setup on a subsequent flight.

4.6.5 Antenna Mode. -

a. Select a radio range station.

b. Tune AN/ARN-89( ) to selected radio range station.

c. Place COMP-ANT-LOOP selector switch in the ANT position.

d. Check that signal strength on the TUNE meter is independent of the aircraft bearing.

e. Check AUDIO control and tuning facilities for smoothness of operation.

f. Place CW-VOICE-TEST toggle switch in the CW position.

g. Check for detectable bfo tone in the audio system.

4.6.6 Compass Mode. -

a. Place COMP-ANT-LOOP selector switch in the COMP position.

b. Place CW-VOICE-TEST toggle switch in the CW position.

c. Check that tone is audible on the station signal.

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d. Fly a steady course towards the station and note bearing, indicator indication, then place CW-VOICE-TEST toggle switch in the TEST position. Bearing indicator indication must be 180 ± 5° displaced from original bearing.

e. Place CW-VOICE-TEST toggle switch in the VOICE position. Bearing indicator must indicate original bearing ±5°.

4.6.7 Zero Compensation Procedure. -

a. Adjust the Homing/ADF indicator such that the compass card indicates 0° at the top pointer.

b. Set switch on the CN-1404/ARN-89 to position "0".

c. Adjust LOOP L-R switch so that goniometer indication is zero degrees.

d. Observe that HOMING/ADF indicator needle reads zero degrees ±2° (top of indicator).

e. Adjust LOOP L-R switch so that goniometer indicates 15, 45, 75, 105, 135, 165, 195, 225, 255, 285, 315, 345 and at each setting, observe that the HOMING/ADF indicator reads the same as the goniometer indication within ±2°. If HOMING/ADF indicator does not read the same as the goniometer indication within ±2°, adjust the compensation screw that the arrow points at until the reading is correct.

f. If adjustments are performed in e, check all points of e rotating goniometer dial in opposite direction and observe no more than ±2° error.

4.6.8 Compensation Flight Check Test Procedure. - Airborne CN-1404/ARN-89 calibration requires that the aircraft fly over a fixed point at least ten miles from a low-frequency radio range station while holding a fixed heading. The fixed reference point should be a prominent landmark easily recognized on aeronautical charts with the surrounding terrain as flat as possible. The radio range station should be transmitting between 200 and 400 KHz. Choose a combination of reference point and radio range station already known to give accurate ADF measurements, if such are available.

Accurate measurements cannot be made unless the aircraft is in level flight attitude. If possible, the calibration should be performed on a calm day to minimize drift and crab angle problems.

The aircraft distorts the electromagnetic field in such a manner that, at the antenna, the direction of arrival of the received signal is shifted toward the nose in the first and second quadrant and towards the tail in the third and fourth quadrant. The CN-1404/ARN-89 installed in an aircraft between the loop antenna and receiver can be set to compensate for errors introduced by the structures of various aircraft. A switch on the Network (Figure 2) permits correction in 2 degree increments up to 20 degrees in the first and third quadrants and is used as an overall adjustment eliminating the requirement for goniometer adjustments.

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NOTE: Normally when the CN-1404/ARN-89 is used for error corrections, the goniometer must have been "zeroed" in accordance with the procedure of paragraph 4.6.7 and does not require further adjustments.

a. The magnetic bearing of the radio station from the flight reference point must be accurately determined and recorded in Column B, Figure 7. Refer to Figure 8 for explanation of ADF bearing terminology.

b. Fly over the reference point at the first aircraft magnetic heading, Column C, Figure 7. At the moment the aircraft is directly over the reference point, record the indicated magnetic bearing to the station (the bearing on the compass dial indicated by the ADF pointer) in the appropriate section of Column D, Figure 7 (CCW when flying left hand turns).

c. Continue to fly straight for a short time. Then make a broad turn and fly over the reference point with the second aircraft magnetic heading. Again, record the indicated magnetic bearing to station in Column D in Figure 7.

d. Repeat Step c for each of the remaining aircraft magnetic headings. A typical flight pattern of an aircraft performing this test is shown in Figure 9.

e. Plot the indicated bearing error data previously obtained from Figure 7 on Figure 10 which is Error (Column E) versus Station Relative Bearing (Column F_T or F_B).

f. After making a plot, average the absolute value of the maximum errors in each quadrant (0° - 90°, 90° - 180°, etc.) and note the value.

g. On the CN-1404/ARN-89, set the COMPENSATION switch to a position corresponding to the average error obtained in f above. Each position of the switch (starting from zero) is equivalent to 2 degrees change in indicated bearing.

h. After making the CN-1404/ARN-89 adjustment, a repeat flight check shall be made and the error versus station relative bearing plotted on Figure 10 once again.

i. Repeat steps f through h if additional compensation is required.

j. Repeat the entire procedure using a second radio range station whose bearing is different than those used to make the compensation and collect data for columns B, thru D, Figure 7, on the new station. The magnetic readings of Column D of Figure 7 must be within $\pm 5°$ of the station magnetic bearing of Column B.

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4.6.9 Compass Range Performance. - Compass Range Performance is defined as the distance at which the bearing indicator needle fails to return to within 5 degrees of original heading when the loop is manually slewed first clockwise and then counter-clockwise. This distance must be at least 50 nautical miles when using an "H" class non-directional radio beacon. The steadiness of the compass indication can be affected by nearby thunderstorms, reflections from nearby mountains, and by skywave interference. The performance is determined by flying directly away from the radio station while slewing the loop as indicated above (compass mode) and noting the bearing. Several such slewing operations should be made in order to establish an average figure. Note that identification of the signal source is extremely important and that the receiver will give good compass indication for greater distances than the identification can ordinarily be distinguished in the COMPASS Mode. It is, therefore, good practice to switch to ANT Mode, adjust the gain manually, determine the identification and then switch back to COMPASS Mode.

4.6.10 COMPASS Performance (over station indicator). - Fly directly towards the radio range station and over it, noting the location of the aircraft with respect to the range station at the start and at the completion of the swing of the bearing indicator needle. The reversal should fall within an inverted cone of 45 degrees with the range station at the apex. Failure to do so is most likely caused by a sense antenna that does not meet the requirements for producing zero signal when directly over the radio range station. A properly located balanced T sense antenna assures proper operation in this respect.

4.7 Production Flight Test. - Production flight tests shall consist of the tests of paragraphs 4.6.8 thru 4.6.10 of this specification.

5. PACKAGING

5.1 This section is not applicable to this specification.

6. NOTES

6.1 Intended Use. - The CN-1404/ARN-89 is to be used with the AN/ARN-89( ) Direction Finder Set. It is meant to be an aid to maintenance personnel in that once the unit is set, it is left in the aircraft. All new or repaired receivers will be shipped with their goniometers "zero compensated". When AN/ARN-89( ) receivers are replaced no goniometer adjustment or flight compensation will be necessary.

6.2 Goniometer Adjustment. - Whenever the CN-1404/ARN-89 is installed, the AN/ARN-89( ) goniometer must be electronically compensated for zero correction over the entire 360° range.

CUSTODIAN

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PREPARING ACTIVITY

Army - EL

Project Number - 5826-A137

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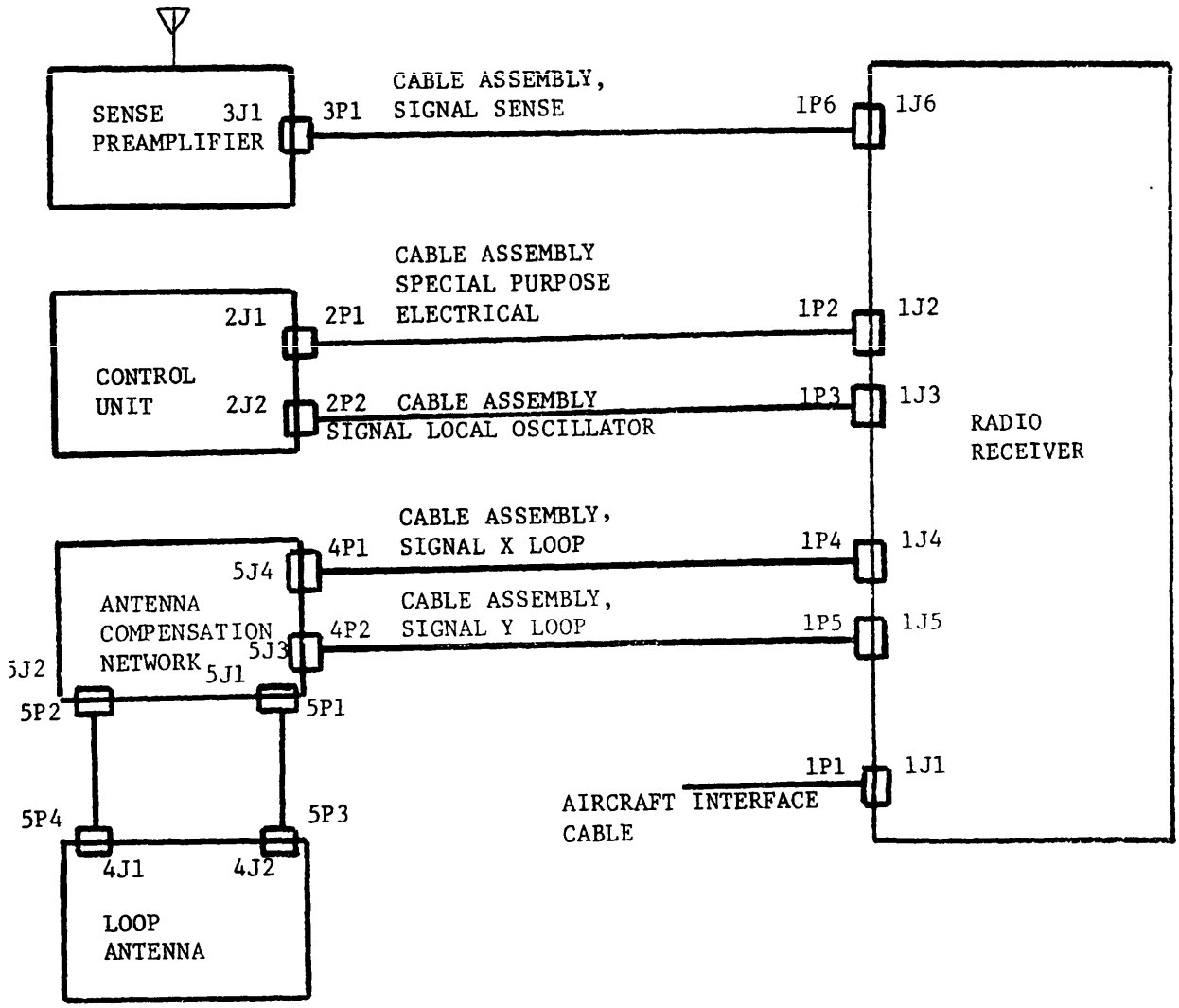


FIGURE 1 CABLING DIAGRAM FOR AN/ARN-89( ) AND CN1404/ARN-89 ANTENNA COMPENSATION NETWORK

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NOTES

1. ALL DIMENSIONS IN INCHES
2. UNIT WEIGHT 0.5 LBS MAX.
3. HEAT DISSIPATION 1.2 WATT MAX.
4. ALL DIMENSIONS ARE MAXIMUM

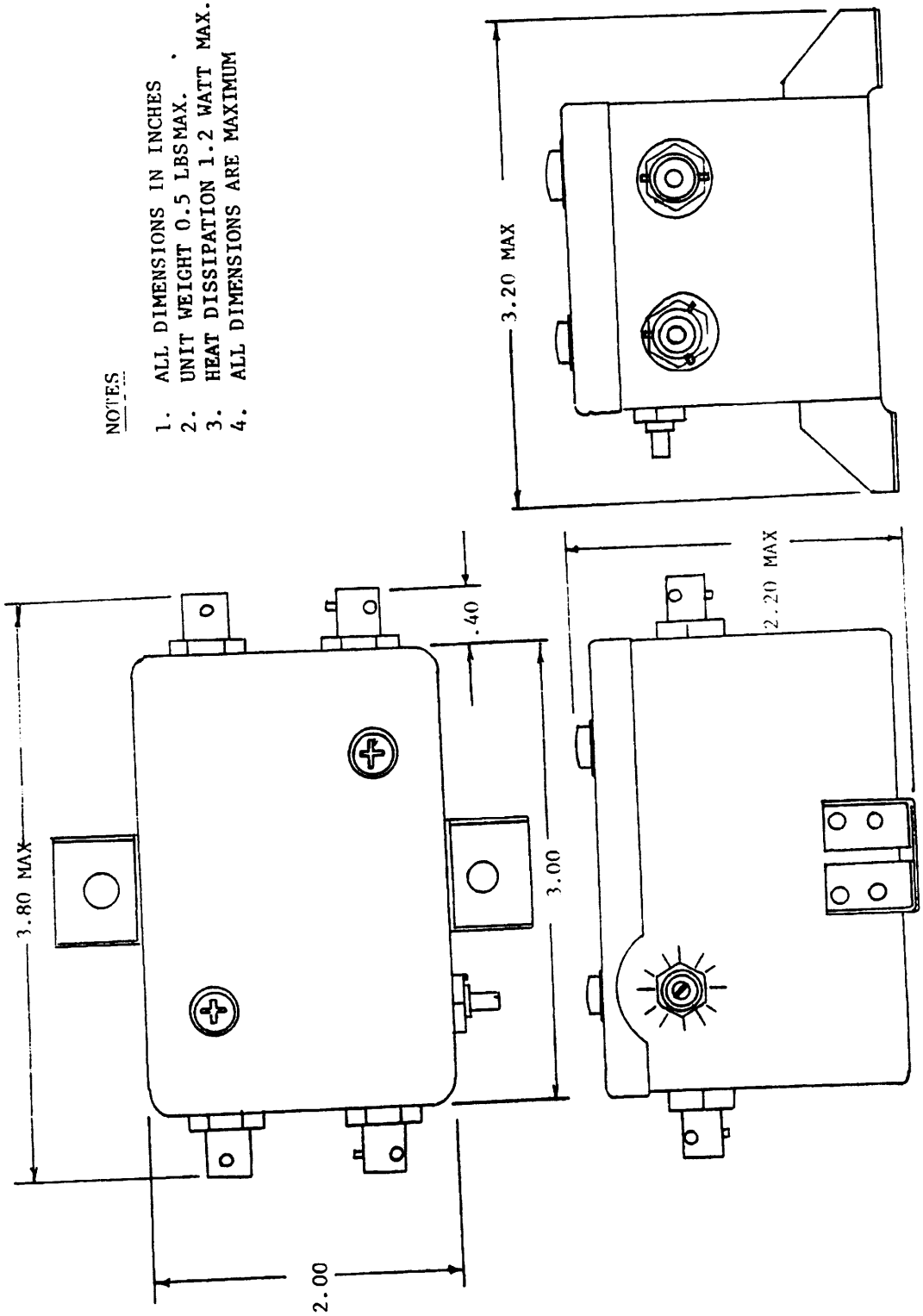


FIGURE 2 OUTLINE DIMENSIONS OF CN-1404/ARN-89

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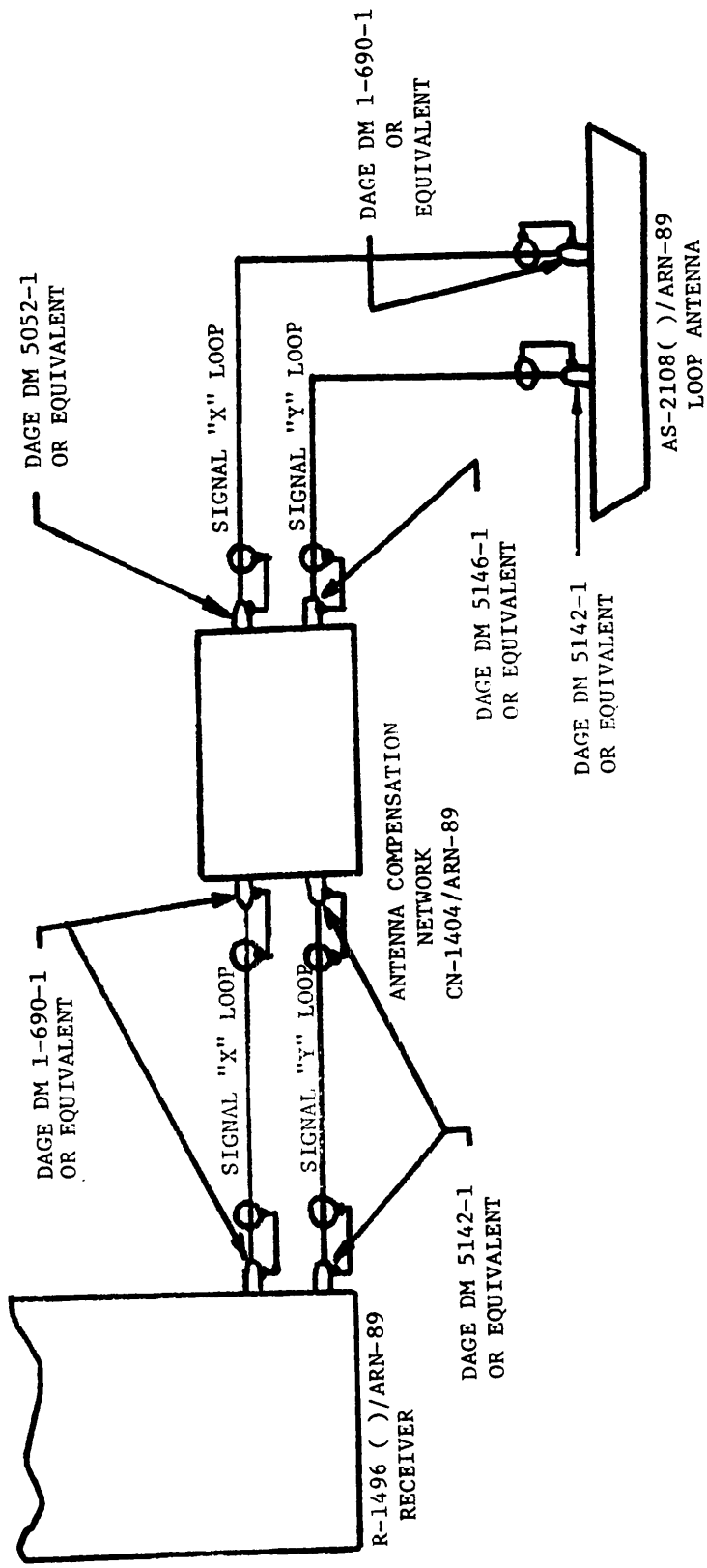


FIGURE 3 INSTALLATION DATA

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NOTES:
1. FOR CABLE LENGTH SEE AIRCRAFT DETAIL SPECIFICATION.
2. CABLE-RG-188A/U

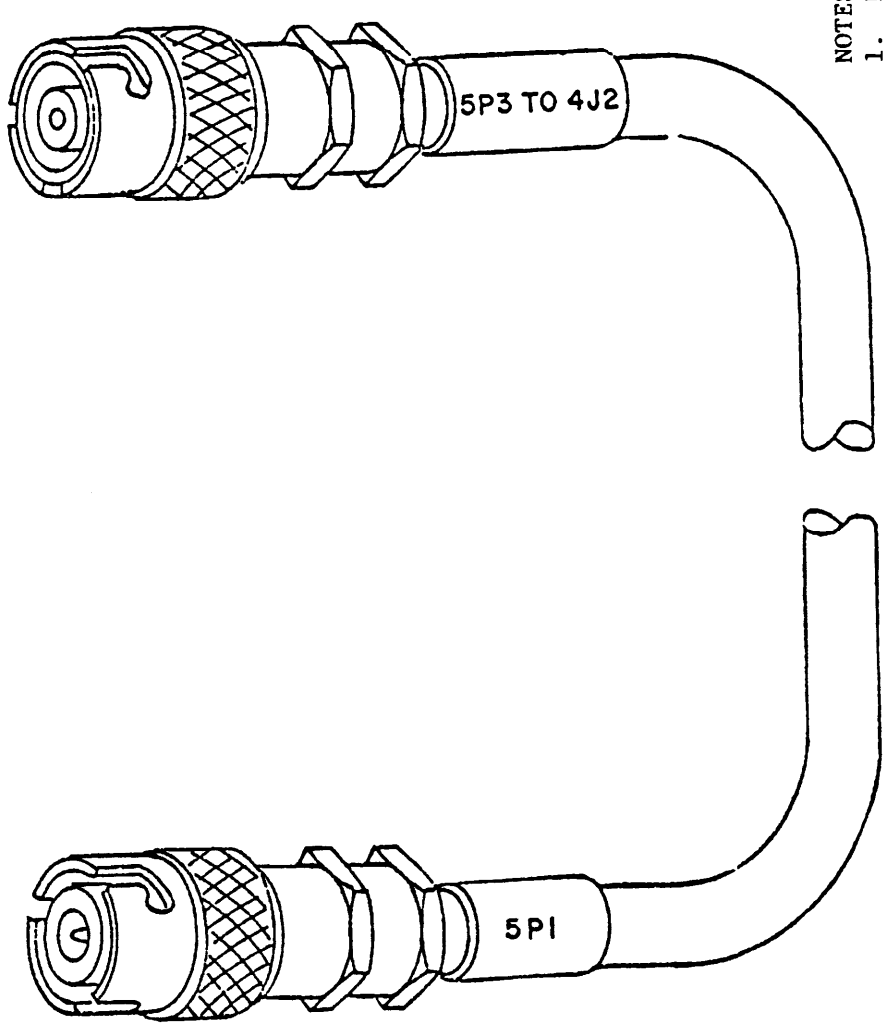


FIGURE 4 "X"- CABLE (CN1404 TO LOOP)

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NOTES:
1. FOR CABLE LENGTH SEE AIRCRAFT DETAIL SPECIFICATION.
2. CABLE-RG 188A/U

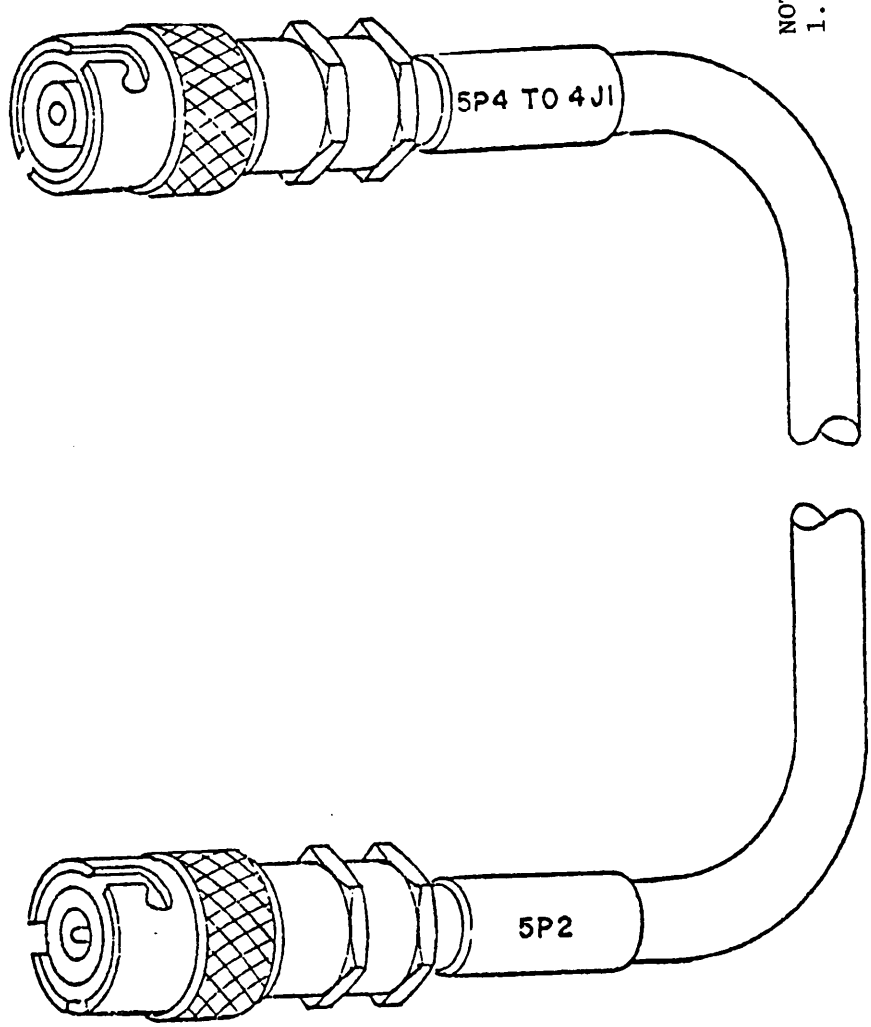
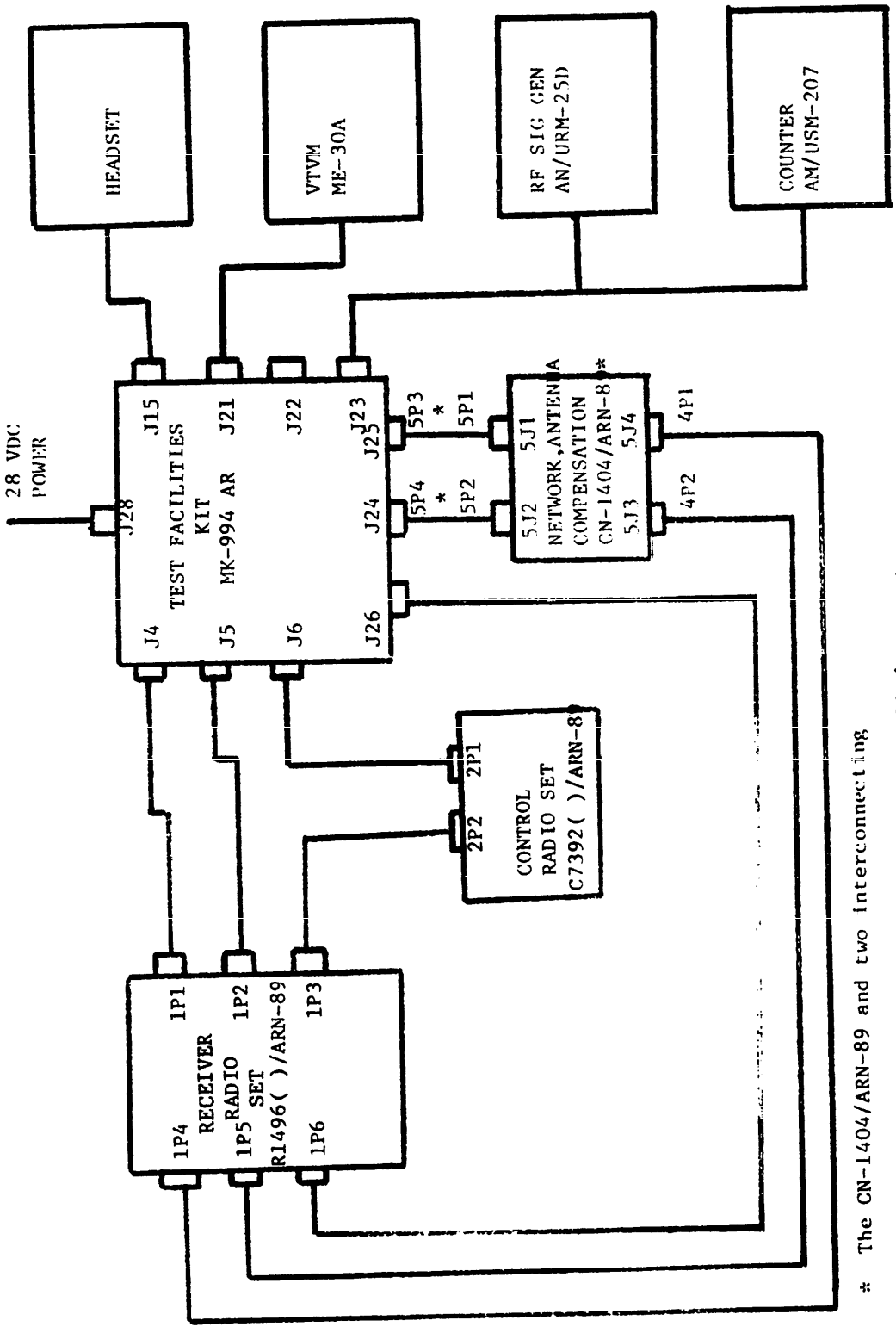


FIGURE 5 "Y" - CABLE (CN1404 TO LOOP)

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* The CN-1404/ARN-89 and two interconnecting cables are auxiliary items and are not supplied as part of the AN/ARN-89( ) System.

FIGURE 6 BENCH TEST SETUP FOR NETWORK, ANTENNA COMPENSATION CN-1404/ARN-89

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DATA FOR LOOP COMPENSATION

Station Used..... Pilot.....

Frequency..... Recorder.....

Reference Point..... Aircraft Type.....

Date..... Aircraft No.....

A Aircraft Relative Heading	B Station Magnetic Bearing	C Aircraft Magnetic Heading (A + B)	D Station/ADF Magnetic Reading		E Error (D-B)	F Station Relative Bearing	
			CW	CCW		F_T	F_B
0*						360*	180*
15						345	165
30*						330*	150*
45						315	135
60*						300*	120*
75						285	105
90*						270*	90*
105						255	75
120*						240*	60*
135						225	45
150*						210*	30*
165						195	15
180*						180*	0*
195						165	345
210*						150*	330*
225						135	315
240*						120*	300*
255						105	285
270*						90*	270*
285						75	255
300*						60*	240*
315						45	225
330*						30*	210*
345						15	195
360*						0*	180*

*READINGS FOR THESE BEARINGS ARE NECESSARY TO DEVELOP A LINEAR COMPENSATION CURVE THOUGH COMPENSATION SCREWS ARE NOT AVAILABLE FOR THESE SETTINGS.

FIGURE 7 LOOP COMPENSATION DATA SHEET

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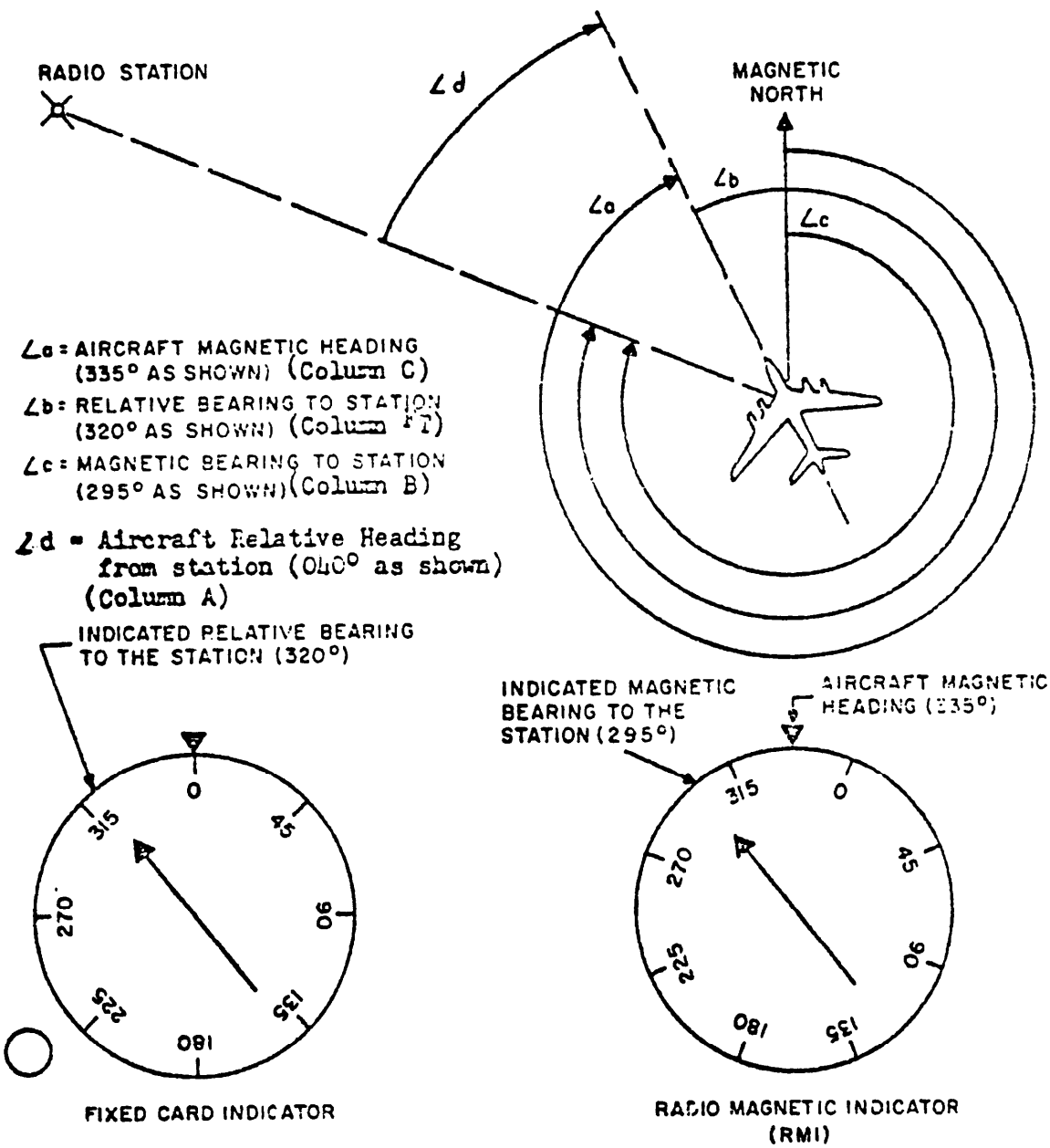
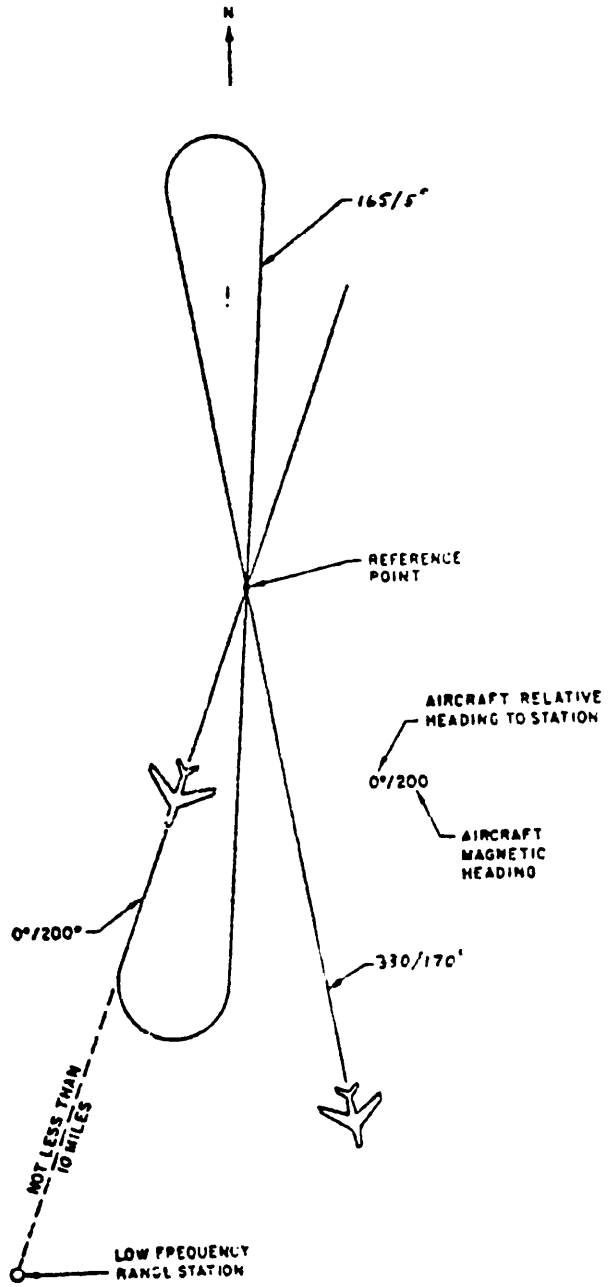


FIGURE 8

ADF BEARING TERMINOLOGY

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AIRCRAFT RELATIVE HEADINGS

WHEN FLYING RT TURNS	WHEN FLYING LFT TURNS
0	0
195	165
30	330
225	135
60	300
255	105
90	270
285	75
120	240
315	45
150	210
345	15
180	180
15	345
210	150
45	315
240	120
75	285
270	90
105	295
300	60
135	225
330	30
165	195
0	0

FIGURE 9 TYPICAL ADF SYSTEM CALIBRATION PATTERN

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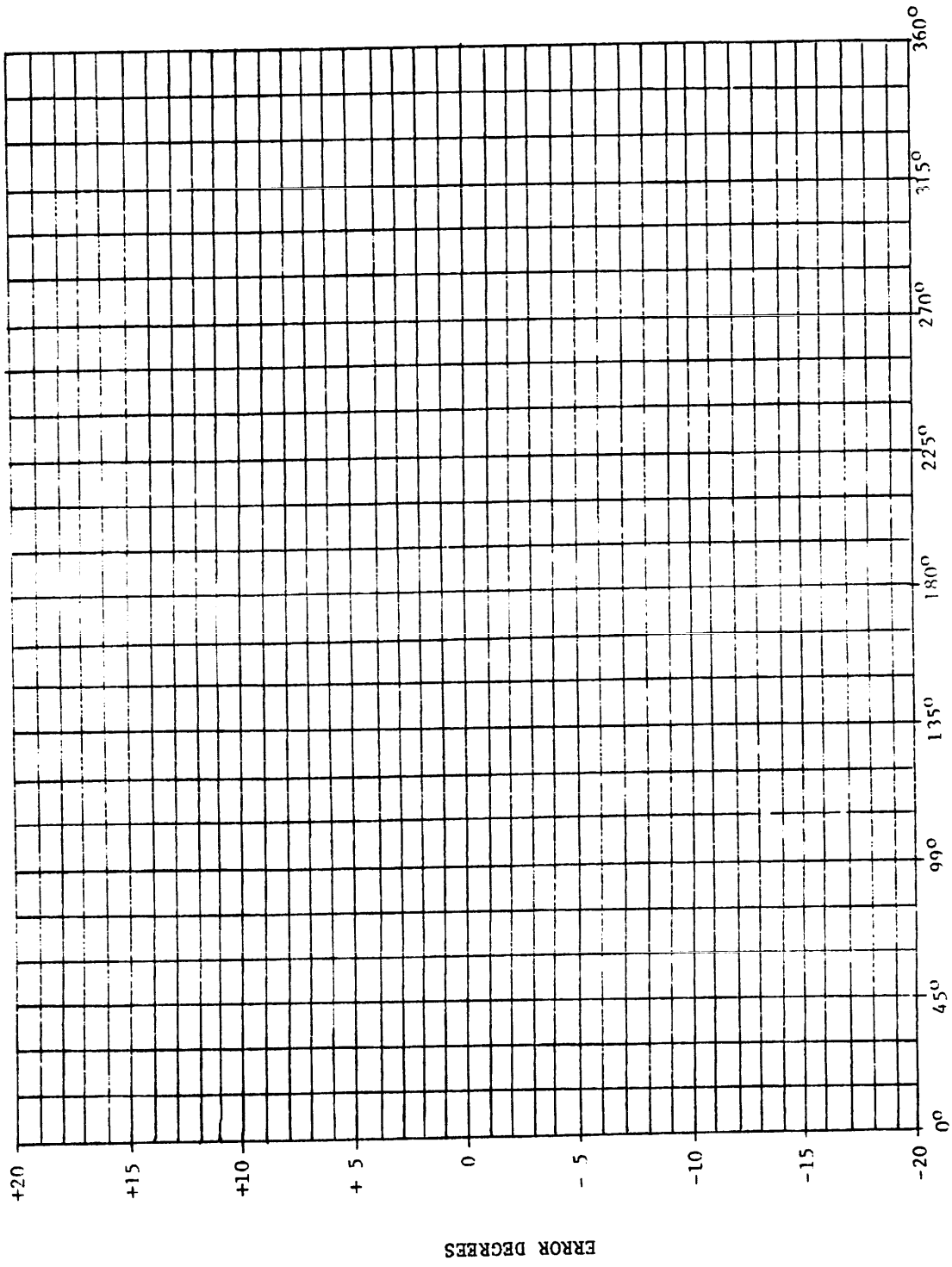


FIGURE 10 Indicated Bearing Error Versus Relative Bearing

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SPECIFICATION ANALYSIS SHEET

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SPECIFICATION MIL-I-49105(EL) INSTALLATION AND ACCEPTANCE TESTING OF NETWORK, ANTENNA COMPENSATION, CN-1404/ARN-89, IN ARMY AIRCRAFT

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

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